

$$C = (A \cup (B \cap C)) \cap (A \cup (\overline{B} \cap \overline{C})) - (A \cap ((B \cap C) \cap (\overline{B} \cap \overline{C}))) \quad (\text{1st Distributive Law})$$

$$F = ((A \cup B) \cap C) \cap ((A \cup \overline{B}) \cap \overline{C}) - (A \cap ((B \cap C) \cap (\overline{B} \cap \overline{C}))) \quad (\text{1st Associative Law})$$

A	B	C	$B \oplus C$	$A \oplus (B \oplus C)$	$(A \oplus B)$	$(A \oplus B) \oplus C$
0	0	0	0	0	0	0
0	0	1	1	1	0	1
0	1	0	1	1	1	0
0	1	1	0	0	1	1
1	0	0	0	0	0	0
1	0	1	1	1	0	1
1	1	0	1	0	1	0
1	1	1	0	1	0	1

47) Suppose that A, B and C are sets such that $A \oplus C = B \oplus C$. Must it be the case that $A = B$?

Solution: $A \oplus C = (A - C) \cup (C - A) \quad (\text{Prob 42})$
 $= (A \cap \overline{C}) \cup (C \cap \overline{A}) \quad (\text{Prob 21(a)})$

$B \oplus C = (B \cap \overline{C}) \cup (C \cap \overline{B}) \quad (\text{Similarly})$

Given $A \oplus C = B \oplus C \therefore (A \cap \overline{C}) \cup (C \cap \overline{A}) = (B \cap \overline{C}) \cup (C \cap \overline{B})$

$\therefore (A \cap \overline{C}) \cup (C \cap \overline{A}) \subseteq (B \cap \overline{C}) \cup (C \cap \overline{B})$ and $(B \cap \overline{C}) \cup (C \cap \overline{B}) \subseteq (A \cap \overline{C}) \cup (C \cap \overline{A})$.

Let, x be an arbitrary element of $(A \cap \overline{C}) \cup (C \cap \overline{A})$, i.e. $x \in (A \cap \overline{C}) \cup (C \cap \overline{A})$.

By Defn. 1, $x \in (A \cap \overline{C}) \vee x \in (C \cap \overline{A})$. By Defn. of OR, at least one of the propositions must be True.

Let, $x \in (A \cap \overline{C})$. By Defn. 2, $x \in A \wedge x \in \overline{C}$. By Defn. of AND, $x \in A$, $x \in \overline{C}$.

By Defn. 5, $x \notin C$. By Defn. of not belongs to, $\neg(x \in C)$. By Defn. of negation, $\neg(\neg(x \in C))$ is False. By Double Negation Law of Propositions,

$x \in C$ is False. $\therefore$ By Defn. of AND, $x \in C \wedge x \in \overline{B}$ is False. By Defn. 2, $x \in C \cap \overline{B}$ is False. By Defn. 3 from Sec-2.1, $x \in (B \cap \overline{C}) \cup (C \cap \overline{B})$.

By Defn. 1, $x \in (B \cap \overline{C}) \vee x \in (C \cap \overline{B})$. By Defn. of OR, $x \in (B \cap \overline{C})$.

By Defn. of intersection, $x \in B \wedge x \in \overline{C}$. By Defn. of AND, $x \in B$ and $x \in \overline{C}$. By Defn. 3 from Sec-2.1, $A \subseteq B$.